Counting more women's work: unpaid helpers, subsidiary activity and the rise in women's employment in India, 2017–18 to 2023–24

Preet Deep Singh*

Abstract

India's Periodic Labour Force Survey shows the worker-population ratio of women aged 15 and over rising from 22.0 per cent in 2017–18 to 40.3 per cent in 2023–24. Using the unit-level data for both years, I reproduce the published estimates and decompose this 18.4-point rise. Own-account work and unpaid help in household enterprises account for 86 per cent of it, agriculture for 73 per cent and women working only in a subsidiary capacity for 48 per cent. Changes in the composition of the population by age, education, marital status, sector and state explain none of it. Weekly hours worked per woman rose by 40 per cent, half as fast as the ratio. In 2023–24, 42 per cent of self-employed women in primary production produced only or mainly for household consumption. The rise is concentrated in the forms of work closest to the boundary of what surveys count as work.

Keywords: women's employment; labour force participation; unpaid family workers; measurement of work; own-use production; India

*Independent Researcher, New Delhi, India. ORCID: 0000-0003-3721-7548. Email: preetds@iima.ac.in.

1 Introduction

For more than a decade, research on women's employment in India was about decline. Successive rounds of the National Sample Survey recorded falling participation of women in the labour force between 2004–05 and 2011–12, despite rapid economic growth, falling fertility and rising education (Afridi, Dinkelman, and Mahajan, 2018; Andres et al., 2017; Chatterjee, Desai, and Vanneman, 2018; Desai and Joshi, 2019; Klasen and Pieters, 2015; Mehrotra and Parida, 2017). The Periodic Labour Force Survey (PLFS), which replaced the quinquennial employment rounds in 2017–18, initially confirmed a low level: the worker-population ratio (WPR) of women aged 15 and over in usual status was 22.0 per cent in 2017–18. Six years later, the same survey reported 40.3 per cent, and the published series rose in every year in between (National Sample Survey Office, 2024).

A rise of 18.4 percentage points in six years is large, and it has prompted a debate about its meaning. Studies using the published tables and unit-level data show that the rise is concentrated among rural women and in self-employment (Ara and Shrivastav, 2025). One interpretation is that it reflects distress, with women entering low-productivity family work as household incomes came under pressure (Jha, Ghatak, and Singh, 2026); another is that additional family labour in agriculture is productive and therefore a positive development (Goldar and Aggarwal, 2026). A third possibility has received less systematic attention: that part of the rise reflects how much of women's existing activity is captured as work. Women's work in rural India is often part-time, seasonal, home-based and close to the line between production for the market, production for the household's own use and domestic work, and surveys are known to be sensitive to how such work is probed (Bardasi et al., 2011; Beneria, 1981; Deshpande and Kabeer, 2024; Hirway and Jose, 2011; Langsten and Salen, 2008).

This paper does not try to choose between these interpretations directly. Instead, it provides the accounting that any interpretation needs. Using the unit-level data of PLFS 2017–18 and 2023–24, I first reproduce the published headline estimates for both years. I then decompose the change in women's WPR by status in employment, by whether women work in a principal or only in a subsidiary capacity, by industry and by rural and urban sector, with design-based standard errors. I compare the distribution of women by principal activity in the two years,

measure what the rise means in hours of work, use a new question in the 2023–24 survey on the destination of output to assess how much of the employment is production mainly for the household's own consumption, and test whether changes in population composition account for the rise.

Five findings stand out. First, own-account work and unpaid help in household enterprises account for 86 per cent of the rise, and regular wage employment for 10 per cent. Second, 48 per cent of the rise consists of women whose principal activity is outside the labour force, typically domestic duties, and who are counted as workers because of a subsidiary economic activity of at least 30 days in the year; 73 per cent of the rise is in agriculture. Third, the rise is much smaller when measured in hours: the mean weekly hours worked per woman in the population rose by 40 per cent, against 84 per cent for the WPR, and the share of women workers who worked fewer than 20 hours in the reference week rose from 20.2 to 32.0 per cent. Fourth, in 2023–24, 42 per cent of self-employed women in primary production reported that their output was only or mainly for the household's own consumption; excluding such work, as the 2013 international standard on statistics of work would (International Labour Organization, 2013), lowers the 2023–24 WPR of women from 40.3 to between 31.5 and 37.0 per cent. Fifth, a shift-share decomposition shows that changes in age, education, marital status, sector and state composition do not explain the rise; it occurred within these groups, and it was largest among rural women aged 35 to 59, women with little schooling and Scheduled Tribe women.

I also document that the field instructions for the 2023–24 survey contain an explicit list prompting enumerators to probe for work in kitchen gardens, orchards, household poultry and dairy and free collection, and worked examples on when such activity makes a woman engaged in domestic duties a subsidiary worker (National Sample Survey Office, 2023a). These were not in the 2017–18 instructions (National Sample Survey Office, 2016), and the survey's own note dates the worked examples to clarifications first provided for PLFS 2022–23 (National Sample Survey Office, 2023b). The definitions of work did not change, and the data cannot show how much the instructions contributed. The point is narrower: the categories in which the rise is concentrated are exactly those in which responses depend most

on probing, so the rise should be read with that in mind.

The paper contributes to three literatures. It adds a detailed accounting of the recent reversal to work on India's women's employment, which has largely studied the earlier decline (Afridi, Dinkelman, and Mahajan, 2018; Klasen and Pieters, 2015; Lahoti and Swaminathan, 2016; Mehrotra and Parida, 2017; Sarkar, Sahoo, and Klasen, 2019). It adds an unusual large-scale case to the literature on the measurement of women's work (Bardasi et al., 2011; Beneria, 1981; Langsten and Salen, 2008), in which a large change in the measured rate is concentrated in the most measurement-sensitive categories. And it uses the destination-of-output question in PLFS to assess the gap between the national definition of work and the 2013 international definition of employment. The analysis is descriptive. It does not attribute the change to any particular cause or policy.

2 Background

2.1 Women's work in India and its measurement

Explanations of the earlier decline in women's participation emphasized rising household incomes and an income effect on women's labour supply, rising enrolment in education, the shortage of jobs considered suitable for women, social norms that restrict women's mobility, and changes in the structure of agricultural work (Afridi, Dinkelman, and Mahajan, 2018; Chatterjee, Desai, and Vanneman, 2018; Jayachandran, 2021; Klasen and Pieters, 2015; Mehrotra and Parida, 2017). The cross-country literature on a U-shaped relation between development and women's participation also predicts that women withdraw from family farm work as household incomes rise and return to the labour force later, through wage employment in services (Gaddis and Klasen, 2014; Goldin, 1994). A rise in women's employment that comes mainly through self-employment in agriculture runs against both of these patterns, which is one reason the recent rise has been questioned (Ara and Shrivastav, 2025; Jha, Ghatak, and Singh, 2026).

Measurement has been a concern throughout. Women's work in family farms and enterprises

is often not reported as work unless respondents are asked about it specifically, and the answers depend on the wording of questions, on the reference period and on who answers (Bardasi et al., 2011; Beneria, 1981; Langsten and Salen, 2008). In India, time-use data have long recorded more economic activity by women than labour force surveys (Hirway and Jose, 2011), and the distinction between work for income, expenditure-saving work for the household and unpaid domestic work is important for understanding women's work (Deshpande and Kabeer, 2024). Earlier panel evidence also shows frequent movements of Indian women into and out of work (Sarkar, Sahoo, and Klasen, 2019), so the measured rate depends on how intermittent work is captured.

2.2 Concepts in the PLFS

The PLFS classifies each person by usual activity status over the 365 days preceding the survey. The principal status is the activity on which a person spent a relatively long time. A person whose principal status is outside the labour force, such as domestic duties, is counted as a worker in a subsidiary capacity if she was engaged in an economic activity for at least 30 days in the year. The headline 'usual status (ps+ss)' measure counts both principal and subsidiary workers. Status in employment distinguishes own-account workers and employers, unpaid helpers in household enterprises, regular wage or salaried employees and casual labourers. A separate current weekly status classifies persons by their activity in the seven days before the survey, and the survey records the hours actually worked on each of those days.

India's national definition of economic activity follows the System of National Accounts production boundary. It includes all production of primary goods, including free collection of uncultivated crops, firewood and fish, whether for sale or for the household's own consumption (National Sample Survey Office, 2016, 2023a). Two codes for persons outside the labour force are relevant: code 92 (attended domestic duties only) and code 93 (attended domestic duties and was also engaged in free collection of goods, sewing, tailoring or weaving for household use). A woman who does domestic work and also collects firewood or vegetables for the household is therefore recorded either as code 93 or, if the collection amounts to 30

days or more in the year, as a subsidiary own-account worker.

The 2013 resolution of the International Conference of Labour Statisticians narrowed the definition of employment to work for pay or profit and classified the production of goods for the household's own final use as a separate form of work, 'own-use production work', outside employment (International Labour Organization, 2013). The PLFS continues to use the earlier, broader concept. In 2023–24, however, the survey asked self-employed persons in primary-sector activities whether their output was for own consumption only, mainly for own consumption, mainly for sale or entirely for sale. This question was not asked in 2017–18.

2.3 Changes in the field instructions

The definitions above are the same in the field instructions for both survey years. The 2023–24 instructions add three elements that the 2017–18 instructions do not contain (Table A1). First, the list of examples of economic activities 'for which special care is needed', which enumerators are told to probe so that such activities are 'not omitted', has a new item: 'Working in kitchen garden, orchards, household poultry, dairy, free collection of fish, small game, wild fruits, vegetables, etc. for household consumption/sale' (National Sample Survey Office, 2023a). Second, a new box of worked examples explains how to distinguish codes 92 and 93 from subsidiary self-employment. In one example, a woman whose main activity is domestic duties and who collects firewood for the household for an hour or two a day for at least 30 days in the year is to be recorded with principal status 92 and a subsidiary economic activity as an own-account worker, not as code 93. Third, a further box applies the same logic to the daily activity record of the current weekly status. The survey's note on updated instructions states that the clarifications in the two boxes of worked examples were first provided for PLFS 2022–23; it does not date the new item in the list (National Sample Survey Office, 2023b). They clarify rather than change the concepts, but they direct attention to the activities that are most likely to be under-reported among women.

3 Data and methods

3.1 Data

I use the unit-level data of the first visit of PLFS 2017–18 and PLFS 2023–24, released by the National Statistics Office of the Ministry of Statistics and Programme Implementation. The first-visit schedules cover the rural and urban samples and are the basis of the official annual estimates in usual status. The files contain 433,339 persons in 102,113 households and 12,773 first-stage units in 2017–18, and 418,159 persons in 101,920 households and 12,743 first-stage units in 2023–24. The analysis sample is persons aged 15 and over: 163,000 women in 2017–18 and 160,042 women in 2023–24, with men used for comparison. The data are anonymized official survey records released for research use; the study involved no contact with participants, and no ethics approval was required for this secondary analysis.

Estimates use the survey multipliers as prescribed in the data documentation: the multiplier is divided by 100 when the sub-sample and the sub-round match and by 200 otherwise, and by the number of quarters to give annual estimates. Standard errors account for the stratified multistage design, with first-stage units within quarter as primary sampling units and the survey strata and sub-strata within state and sector as strata, using the survey package in R (Lumley, 2004). Because the two years are independent samples, the standard error of a change is the square root of the sum of the two squared standard errors. I harmonize state codes across years by combining Jammu and Kashmir with Ladakh and Dadra and Nagar Haveli with Daman and Diu, so that both years refer to the same territories.

3.2 Definitions

A person is a usual-status worker if her principal status is a work code (11–51), or if her principal status is not a work code and she reports a subsidiary economic activity with a work code. Workers are classified by the status of the principal activity for principal workers and of the subsidiary activity for subsidiary-only workers, into own-account workers and employers (codes 11–12), unpaid helpers (21), regular wage or salaried employees (31) and

casual labourers (41, 51). Agriculture is NIC 2008 divisions 01–03. Weekly hours are the hours actually worked in all economic activities on the seven days before the survey. These are the official definitions, and with them the estimates reproduce the published figures (Section 4.1).

3.3 Decompositions

Because the WPR is the sum of the shares of the population in each status, the change in the WPR is the exact sum of the changes in the status-specific ratios, and the same holds for the split into principal and subsidiary-only workers and into agriculture and other industries. To assess whether the rise reflects changes in the composition of the population, I use a Kitagawa decomposition (Kitagawa, 1955). With cells c defined by state, rural or urban sector, age group (15–24, 25–34, 35–44, 45–59, 60+), education (below primary, primary or middle, secondary or higher secondary, graduate or above) and marital status, the change in the WPR is

$$\Delta W = \sum_c \bar{r}_c \, \Delta p_c + \sum_c \bar{p}_c \, \Delta r_c,$$

where p_c is the population share of cell c, r_c its WPR, and bars denote averages of the two years. The first term is the composition component and the second the rate component. Cells present in only one year (0.32 per cent of the population) are excluded from the decomposition and the shares are renormalized.

4 Results

4.1 Reproducing the official estimates

The estimates from the unit-level data reproduce the published figures. For 60 published values (the LFPR, WPR and unemployment rate in usual status and the LFPR and WPR in current weekly status, for men and women, in rural areas, urban areas and the whole country,

in both years), 59 are identical after rounding to one decimal place and the remaining one (the LFPR in current weekly status for men in 2017–18, published as 75.1 and estimated as 75.16) differs by 0.06 points. Table 1 shows the national values. The women's WPR rose from 22.0 to 40.3 per cent, a rise of 18.4 points (95 per cent CI 17.7 to 19.0). Men's WPR rose by 5.1 points, from 71.2 to 76.3 per cent. The women's unemployment rate fell from 5.6 to 3.2 per cent, so almost all of the rise in participation took the form of employment.

Table 1: Published and reproduced headline estimates, persons aged 15 and over, India (per cent)

		2017–18		2023–24	
Indicator	Sex	Published	Estimated	Published	Estimated
LFPR, usual status	Women	23.3	23.29	41.7	41.66
LFPR, usual status	Men	75.8	75.80	78.8	78.80
WPR, usual status	Women	22.0	21.98	40.3	40.34
WPR, usual status	Men	71.2	71.17	76.3	76.28
Unemployment rate, usual status	Women	5.6	5.64	3.2	3.15
Unemployment rate, usual status	Men	6.1	6.10	3.2	3.19
LFPR, current weekly status	Women	21.1	21.08	35.6	35.57
LFPR, current weekly status	Men	75.1	75.16	77.5	77.54
WPR, current weekly status	Women	19.2	19.18	33.8	33.81
WPR, current weekly status	Men	68.6	68.65	73.8	73.78

Published values from the NSO press note of 23 September 2024. Estimated values are computed from the unit-level data with the official weights. 59 of 60 published values (including rural and urban values not shown) are reproduced exactly after rounding.

4.2 Where the rise occurred

Table 2 and Figure 1 decompose the change in the women's WPR. Nationally, own-account work and employers added 8.0 points and unpaid helpers 7.8 points, together 86 per cent of the rise of 18.4 points. Regular wage employment added 1.8 points and casual labour 0.8 points. Agriculture accounts for 13.4 points, or 73 per cent. The rise in principal workers was 9.6 points; the share of women who are workers only in a subsidiary capacity rose almost five-fold, from 2.3 to 11.0 per cent, and accounts for 48 per cent of the rise.

The rise was concentrated in rural areas. Among rural women the WPR rose by 22.9 points, from 23.7 to 46.5 per cent. Unpaid helpers added 10.5 points, of which 6.5 points were subsidiary-only helpers, and own-account workers added 10.1 points, of which 3.5 were

subsidiary-only. Subsidiary-only workers account for 50 per cent of the rural rise and agriculture for 81 per cent. Regular wage employment of rural women rose by 1.2 points, and casual labour in a principal capacity did not change (-0.1 points).

Urban women's WPR rose by a smaller 7.8 points, and the composition was different: regular wage employment rose by 3.4 points, own-account work by 3.1 points and unpaid help by 1.5 points, while casual labour fell slightly. Most of the urban rise was in principal status and outside agriculture.

Men's employment changed in the same direction but on a much smaller scale. Men's WPR rose by 5.1 points; unpaid help rose by 1.9 points and subsidiary-only work by 1.6 points, from 0.5 to 2.1 per cent of men. The shift towards self-employment and subsidiary work therefore affected both sexes, but for women it was large enough to nearly double the measured employment rate.

Table 2: Decomposition of the women's worker-population ratio by status, capacity and industry, women aged 15 and over (per cent of population)

	Rural			Urban			All India		
	2017–18	2023–24	Change (SE)	2017–18	2023–24	Change (SE)	2017–18	2023–24	Change (SE)
Worker-population ratio	23.7	46.5	22.9 (0.46)	18.2	26.0	7.8 (0.45)	22.0	40.3	18.4 (0.35)
By status in employment									
Own-account worker or employer	4.5	14.6	10.1 (0.26)	4.3	7.4	3.1 (0.24)	4.4	12.4	8.0 (0.20)
Unpaid helper	9.1	19.6	10.5 (0.34)	2.0	3.5	1.5 (0.16)	7.0	14.8	7.8 (0.24)
Regular wage or salaried	2.5	3.6	1.2 (0.14)	9.5	12.9	3.4 (0.32)	4.6	6.4	1.8 (0.14)
Casual labour	7.5	8.7	1.2 (0.25)	2.4	2.2	-0.2 (0.16)	6.0	6.7	0.8 (0.18)
By capacity									
Principal status	20.9	32.4	11.5 (0.42)	17.1	22.2	5.1 (0.43)	19.7	29.3	9.6 (0.32)
Subsidiary status only	2.8	14.1	11.4 (0.30)	1.0	3.8	2.7 (0.17)	2.3	11.0	8.8 (0.22)
Own-account, subsidiary only	0.7	4.2	3.5 (0.15)	0.5	1.9	1.4 (0.12)	0.7	3.5	2.8 (0.11)
Unpaid helper, subsidiary only	1.5	8.0	6.5 (0.23)	0.3	1.1	0.8 (0.08)	1.1	5.9	4.8 (0.16)
By industry									
Agriculture	17.3	35.8	18.5 (0.43)	1.6	3.2	1.5 (0.18)	12.5	26.0	13.4 (0.31)
Other industries	6.3	10.8	4.4 (0.25)	16.5	22.8	6.3 (0.43)	9.5	14.4	4.9 (0.23)

Usual status (ps+ss). Components in each panel sum to the worker-population ratio. Agriculture is NIC 2008 divisions 01–03. Standard errors account for the survey design.

4.3 From domestic duties to work

The rise in work was mirrored by a fall in the share of women whose principal activity was domestic duties. Because the two surveys are independent cross-sections, these are net changes in shares, not observed transitions of individual women. Table 3 shows the distribution of women by usual principal activity. Among rural women, the share whose

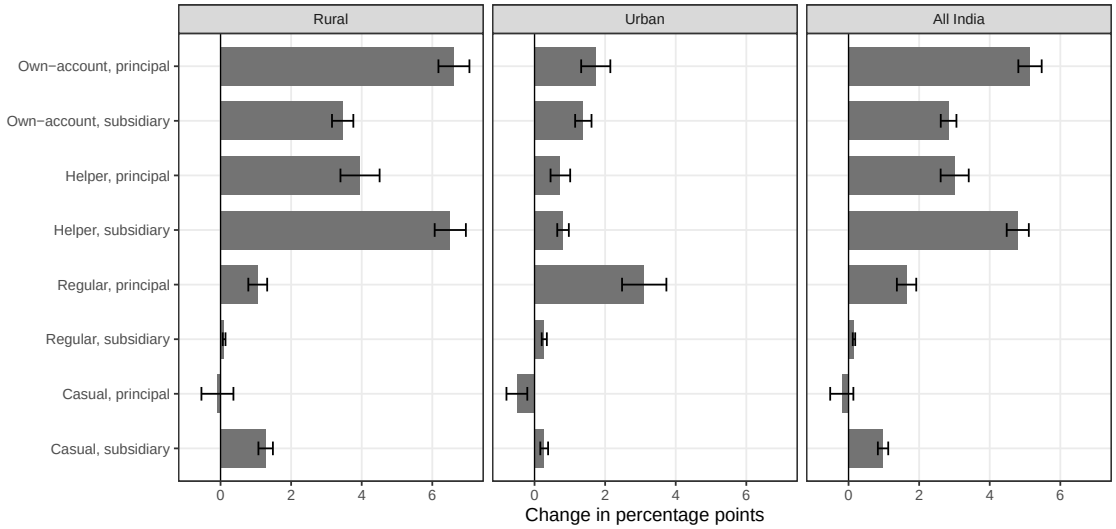


Figure 1: Contributions to the change in women's worker-population ratio, 2017–18 to 2023–24, by status and capacity (percentage points)

principal activity was domestic duties only (code 92) fell from 42.0 to 30.4 per cent, while the share in domestic duties combined with free collection and similar activities for the household (code 93) was almost unchanged (18.4 and 18.0 per cent). Principal-status unpaid helpers rose from 7.7 to 11.6 per cent and principal-status own-account workers from 3.8 to 10.4 per cent. Together, the principal-status work categories of rural women rose by 11.5 points, close to the fall in the domestic-duties-only share. The shares in education and in other categories outside the labour force changed little. In urban areas, too, the domestic-duties-only share fell, from 55.6 to 49.5 per cent.

The rise in subsidiary-only work took place within the domestic-duties categories, since subsidiary-only workers are, by construction, women whose principal activity is outside the labour force. In 2023–24, 44.4 per cent of rural subsidiary-only women workers had domestic duties only (code 92) as their principal status and 45.8 per cent had code 93, proportions similar to 2017–18. What changed is the number of such women: the subsidiary-only WPR of rural women rose from 2.8 to 14.1 per cent. The rise in women's employment therefore consists to a large extent of women who are still mainly engaged in domestic duties but now report, or are recorded as having, a secondary economic activity.

Table 3: Women aged 15 and over by usual principal activity status (per cent)

	Rural			Urban		
Usual principal activity status	2017–18	2023–24	Change	2017–18	2023–24	Change
Own-account worker or employer	3.8	10.4	6.6	3.8	5.5	1.7
Unpaid helper	7.7	11.6	4.0	1.7	2.4	0.7
Regular wage or salaried	2.5	3.5	1.1	9.4	12.5	3.1
Casual labour	7.0	6.9	-0.1	2.2	1.7	-0.5
Unemployed	1.0	1.2	0.3	2.2	2.1	-0.1
Attending education	10.4	10.9	0.5	13.0	12.1	-0.9
Domestic duties only (92)	42.0	30.4	-11.6	55.6	49.5	-6.1
Domestic duties and free collection etc. (93)	18.4	18.0	-0.3	4.7	6.9	2.2
Other, not in labour force	7.4	7.0	-0.4	7.4	7.1	-0.2

Principal status only; women in codes 92, 93, education and other categories may also be subsidiary workers. Columns sum to 100.

4.4 Intensity of work

A worker in usual status may have worked very little in any given week. Table 4 reports, for women who are usual-status workers, the share who were employed in the reference week by current weekly status, mean hours actually worked in the week (including zero for those who did not work) and the share who worked fewer than 20 hours. Among all women workers, the share employed in the reference week fell from 86.2 to 83.7 per cent, mean weekly hours fell from 36.6 to 28.2, and the share working fewer than 20 hours rose from 20.2 to 32.0 per cent.

Part of the fall in average hours reflects the changed composition of workers. Subsidiary-only workers worked few hours in both years: subsidiary-only helpers averaged 12.6 hours in 2017–18 and 10.7 hours in 2023–24, and 76 per cent of them worked fewer than 20 hours in 2023–24. Their number rose several-fold. Hours also fell within the principal-status groups, by 3.4 hours for principal-status helpers and 5.9 hours for principal own-account workers.

A summary measure that is not affected by where the line between workers and non-workers is drawn is the mean weekly hours of work per woman in the population. Because first visits are spread over all four quarters of the survey year, it averages over seasons, although a single reference week can still miss intermittent work. It rose from 8.12 (SE 0.094) to 11.39 (SE 0.094) hours, an increase of 40 per cent. The WPR rose by 84 per cent over the same period. In rural areas, hours per woman rose from 8.15 to 12.37, and in urban areas from

8.05 to 9.13. One caution is that weekly hours per worker also fell for men, from 50.6 to 44.9, and hours per man in the population fell from 36.04 to 34.24. Part of the fall in hours per worker may therefore reflect changes in how hours were reported rather than changes specific to women; if so, the increase in hours per woman may be understated. Even so, the rise in women's hours of work is about half the rise in the share of women counted as workers.

Table 4: Intensity of work among usual-status workers aged 15 and over

	Employed in reference week (%)		Mean hours worked in week		Worked under 20 hours (%)	
	2017–18	2023–24	2017–18	2023–24	2017–18	2023–24
All women workers	86.2	83.7	36.6	28.2	20.2	32.0
Rural	83.7	81.8	34.1	26.6	23.2	34.9
Urban	93.5	91.6	44.0	35.1	11.4	20.0
Own-account, principal	95.0	96.5	40.1	34.2	10.8	15.5
Own-account, subsidiary only	69.7	82.3	17.7	17.2	51.5	55.7
Unpaid helper, principal	85.2	89.4	34.3	30.9	20.4	23.7
Unpaid helper, subsidiary only	48.8	54.4	12.6	10.7	68.6	76.4
Regular wage, principal	98.5	98.3	50.7	42.1	2.4	6.4
Casual labour, principal	84.0	83.0	34.1	29.2	23.4	26.4
Casual labour, subsidiary only	41.7	32.0	11.8	9.2	71.9	76.3
All men workers (comparison)	96.3	96.7	50.6	44.9	5.8	7.7

Hours actually worked in all economic activities in the seven days before the survey, including zero for workers who did not work in that week. Regular wage workers in a subsidiary capacity only are few and not shown.

4.5 Production for the household's own use

The 2023–24 survey asked self-employed persons in primary-sector activities (NIC divisions 01–09) about the destination of their output. Among women own-account workers and helpers in these activities, 16.2 per cent reported that the output was for own consumption only, 25.9 per cent that it was mainly for own consumption with some sale, 51.8 per cent that it was mainly for sale and 6.1 per cent that it was entirely for sale (Table 5). The corresponding share for own consumption only or mainly was 25.5 per cent among men. The question was answered for all of these workers. Production mainly for own use was most common among rural women working in a subsidiary capacity: 52.3 per cent of them produced only or mainly for the household, compared with 36.3 per cent of rural women working in a principal capacity.

Under the 2013 international definition, production of goods intended mainly for the house-

hold's own final use is own-use production work, not employment (International Labour Organization, 2013). Table 5 shows how the 2023–24 WPR would change if such work were excluded. Excluding women whose output was for own consumption only lowers the women's WPR from 40.3 to 37.0 per cent; also excluding those whose output was mainly for own consumption lowers it to 31.5 per cent. For rural women, the corresponding values are 46.5, 41.9 and 34.3 per cent. These are approximations: the classification is by destination of output and applies only to self-employment in primary activities, and some excluded women may have other work that the survey does not record. They also cannot be compared with 2017–18, when the question was not asked. They show, however, that a substantial part of measured employment of women in 2023–24 is production mainly for the household, which the 2013 standard classifies as own-use production work rather than employment.

Table 5: Destination of output of self-employed workers in primary activities, and the worker-population ratio excluding production for own use, 2023–24 (per cent)

Panel A: destination of output	Women, rural, principal	Women, rural, subsidiary only	Women, all	Men, all
Own consumption only	12.8	21.8	16.2	5.1
Mainly own consumption, some sale	23.4	30.5	25.9	20.4
Mainly for sale	56.8	43.3	51.8	66.0
Entirely for sale	6.9	4.4	6.1	8.5
Panel B: WPR, 2023–24	*Women, rural*	*Women, urban*	*Women, all*	*Men, all*
WPR as published definition	46.5	26.0	40.3	76.3
Excluding own consumption only	41.9	25.6	37.0	75.1
Excluding own consumption only or mainly	34.3	25.1	31.5	70.4

Panel A: own-account workers, employers and unpaid helpers in NIC 2008 divisions 01–09, by capacity in which the activity is pursued; columns sum to 100. Panel B: worker-population ratio of persons aged 15 and over, after excluding self-employed workers in these divisions whose output was for own consumption only, or only or mainly. The question was not asked in 2017–18.

4.6 Composition

Women's employment varies strongly with age, education, marital status and place of residence, so part of the change in the WPR could reflect changes in the composition of the population between the two surveys. Table 6 shows that it does not. With 4,093 cells defined by state, sector, age group, education and marital status, the composition component is -1.06 points and the rate component is 19.42 points. Composition slightly lowered the expected ratios of casual labour and unpaid help and slightly raised that of regular wage employment. The composition component is between -1.06 and 0.06 points for all sets of

cells considered, including cells that add social group (Table A2, Panel A). Restricting the population to women of working age (15–59) gives the same picture: their WPR rose by 19.9 points, of which 9.4 points were subsidiary-only work (Table A2, Panel B). The rise in women's employment took place within groups of women, not through a shift towards groups with higher employment.

Table 6: Kitagawa decomposition of the change in women's worker-population ratio, 2017–18 to 2023–24 (percentage points)

	Total change	Composition	Rates within cells
Worker-population ratio	18.36	-1.06	19.42
Own-account worker or employer	7.98	0.01	7.97
Unpaid helper	7.80	-0.55	8.35
Regular wage or salaried	1.80	0.41	1.39
Casual labour	0.79	-0.92	1.71

Cells: state by sector by age group by education by marital status (4,093 cells present in both years). Women aged 15 and over.

4.7 Who and where

The rise was broad-based but uneven (Table 7). Among rural women, the WPR rose by 13.7 points for those aged 15–24 and by 31.6 and 32.6 points for those aged 35–44 and 45–59. It rose more for women with below-primary (26.0 points) or primary or middle schooling (27.5 points) than for graduates (12.1 points). It rose in every quintile of household consumption per person, least in the poorest (17.9 points) and by 25.3 and 23.1 points in the middle and richest quintiles. It rose most among Scheduled Tribe women, from 37.4 to 66.9 per cent. Among the youngest rural women, the rise came mostly through subsidiary work (9.5 of 13.7 points). In urban areas the increases were smaller in every group.

Across the 20 most populous states, the change in women's WPR ranged from 2.3 points in Andhra Pradesh and 5.5 points in Delhi to 35.0 points in Jharkhand and 37.2 points in Assam (Figure 2; Table A3 gives the state values). Across these states, the correlation between the change in the WPR and the change in the combined ratio of own-account workers and helpers is 0.97. States differed in which of the two categories rose: in Jharkhand, Odisha and Madhya Pradesh it was mainly unpaid helpers, in Assam and West Bengal mainly

Table 7: Women's worker-population ratio by group, women aged 15 and over (per cent)

	Rural			Urban		
	2017–18	2023–24	Change	2017–18	2023–24	Change
Age group						
15–24	9.3	23.0	13.7	8.4	13.8	5.3
25–34	27.2	48.0	20.8	22.9	30.6	7.7
35–44	36.0	67.5	31.6	27.6	38.7	11.1
45–59	33.9	66.5	32.6	22.1	32.4	10.4
60+	11.7	27.1	15.4	7.4	10.2	2.7
Education						
Below primary	29.1	55.2	26.0	22.0	30.3	8.3
Primary or middle	20.8	48.3	27.5	16.2	26.7	10.5
Secondary or higher secondary	14.8	32.4	17.6	11.1	17.2	6.1
Graduate or above	20.9	33.0	12.1	26.7	32.5	5.9
Household consumption quintile						
Q1 (poorest)	24.8	42.7	17.9	17.2	24.4	7.3
Q2	22.8	45.3	22.4	17.7	25.9	8.2
Q3	23.1	48.3	25.3	18.2	25.6	7.4
Q4	24.2	49.3	25.0	18.1	26.3	8.2
Q5 (richest)	23.4	46.5	23.1	19.5	27.4	8.0
Social group						
Scheduled Tribe	37.4	66.9	29.5	22.4	36.3	13.9
Scheduled Caste	24.2	46.2	22.0	22.9	30.0	7.2
Other Backward Class	22.8	45.8	23.0	18.5	25.4	6.8
Others	18.3	36.7	18.4	15.8	24.1	8.2

Usual status (ps+ss). Consumption quintiles are of monthly household consumption expenditure per person, within year and sector.

own-account workers. Because self-employment accounts for most of the national rise, this correlation is partly mechanical. The exceptions are states with smaller rises in which wage work accounted for half or more of the change, such as Kerala, Tamil Nadu and Delhi (Table A3). Elsewhere the pattern of a rise concentrated in self-employment holds, so the national result does not reflect a few states with unusual estimates.

5 Discussion

The accounting establishes four facts about the measured rise in Indian women's employment between 2017–18 and 2023–24. It was a rise in self-employment, mostly in agriculture and mostly in rural areas. Nearly half of it consisted of women whose main activity remained domestic duties and who were counted as workers through a subsidiary activity. It amounted to a much smaller increase in hours of work than in the number of workers. And a large

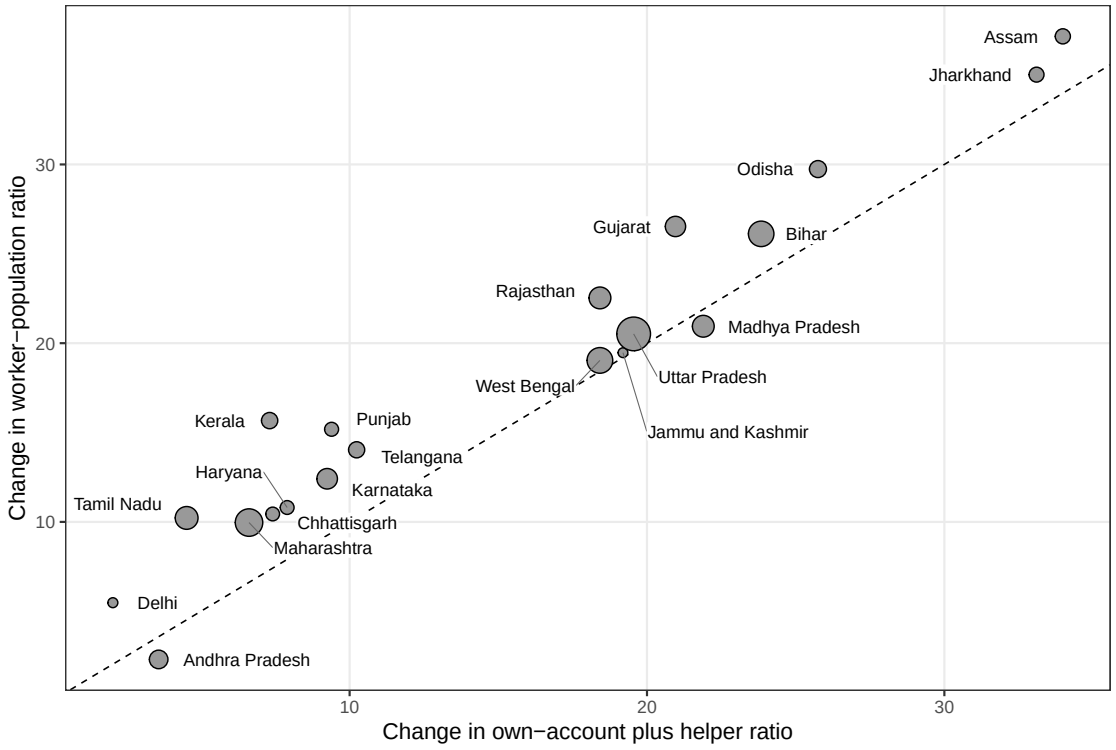


Figure 2: Change in women's worker-population ratio and in the self-employed (own-account plus helper) ratio, 20 most populous states, 2017–18 to 2023–24 (percentage points)

share of the self-employment of women in primary production in 2023–24 was production mainly for the household's own consumption.

These facts are consistent with more than one interpretation, and the data cannot separate them. If more women took up or expanded work on family farms and livestock because of pressure on household incomes, the rise would look like this (Jha, Ghatak, and Singh, 2026). If additional family labour reflects productive opportunities in agriculture and allied activities, the pattern would also look like this (Goldar and Aggarwal, 2026). And if the same activities were recorded more completely, after field instructions that direct enumerators to probe for kitchen gardens, poultry, dairy and free collection and explain when such activity makes a woman engaged in domestic duties a subsidiary worker, the rise would again look like this. The categories that rose are the ones most sensitive to probing in the measurement literature (Bardasi et al., 2011; Langsten and Salen, 2008): unpaid family work, subsidiary activity, small-scale livestock and collection, and production for own use.

Some patterns bear on the relative importance of these explanations without settling it. The worked examples in the instructions describe women who would otherwise be coded 93 being recorded instead with principal status 92 and a subsidiary activity; such a reclassification would, other things equal, lower the share in code 93. That share hardly changed among rural women and rose among urban women, while the share in code 92 fell, mainly because more women reported a principal economic activity. The rise was also gradual: in the published series, the women's WPR was 22.0, 23.3, 28.7, 31.4, 31.7, 35.9 and 40.3 per cent in the seven years from 2017–18 to 2023–24 (National Sample Survey Office, 2024), so about half of the rise had occurred by 2021–22, before the worked examples that the survey's note dates to 2022–23. These patterns are suggestive rather than diagnostic. And men also show a rise in unpaid help and subsidiary work, on a smaller scale. A full assessment would need the annual unit-level data for all years and, ideally, an experimental comparison of probing versions within the survey.

Whatever the mix of causes, the findings have implications for how the rise is used. Measured as hours, the increase in women's work is about half as large as measured by the WPR. Measured against the 2013 international definition of employment, the women's WPR in 2023–24

would be between 31.5 and 37.0 per cent rather than 40.3 per cent. Comparisons of India's women's employment with other countries, or with earlier Indian estimates, should therefore use comparable measures: hours, principal status, or employment excluding own-use production. The PLFS already contains the information needed to produce such measures. Publishing them alongside the headline rates, and reporting principal and subsidiary workers separately, would make the series easier to interpret.

The analysis has limitations. It compares two years six years apart, and the period includes the COVID-19 pandemic; the end points do not show the path. The decomposition is an accounting, not a causal analysis. The destination-of-output question is available only for 2023–24, so the own-use adjustment cannot be applied to 2017–18. The fall in reported hours for men suggests that hours may not be fully comparable between the years. And the assessment of changes in field instructions is based on the published instructions; the data cannot show how they were applied.

6 Conclusion

The rise in women's employment in India between 2017–18 and 2023–24 is reproduced, to within rounding, by the unit-level data, and it is real in the sense that the survey records many more women as workers under formally unchanged definitions, although the guidance to enumerators changed. It is concentrated in own-account work and unpaid help in agriculture, in rural areas, among women who work in a subsidiary capacity alongside domestic duties, and in production often mainly for the household's own use. It is not explained by changes in the composition of the population, and in hours of work it is about half as large. The rise therefore lies largely in forms of work close to the boundary of what surveys count as work. Whether it reflects new work, more work by women already working, or more complete recording of existing work matters for how it is read, and answering that question will require the measures this paper proposes to be tracked over time.

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Preet Deep Singh is an Independent Researcher, New Delhi, India. Please direct all correspondence to preetds@iima.ac.in. ORCID: 0000-0003-3721-7548.

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Data availability. The unit-level data of the Periodic Labour Force Survey 2017–18 and 2023–24 are produced by the National Statistics Office, Ministry of Statistics and Programme Implementation, Government of India, and are available from the Ministry's microdata portal under its terms of use. The analysis code is available from the author on request and will be deposited in a public repository on acceptance.

Ethics. The study is a secondary analysis of anonymized official survey data released for research use. It involved no contact with human participants, and no ethics approval was required.

Declaration of generative AI use. Claude (Anthropic) and other generative AI tools were used to help write and check the analysis code, check the references and draft and edit the text of the manuscript, under the author's direction. The author reviewed and verified all output, including every number, reference and statement, and takes full responsibility for the content.

References

Afridi, F., Dinkelman, T., and Mahajan, K. (2018). Why are fewer married women joining the work force in rural India? A decomposition analysis over two decades. *Journal of Population Economics*, *31*(3), 783–818. https://doi.org/10.1007/s00148-017-0671-y

Andres, L. A., Dasgupta, B., Joseph, G., Abraham, V., and Correia, M. (2017). *Precarious drop: Reassessing patterns of female labor force participation in India* (Policy Research Working Paper 8024). World Bank. https://doi.org/10.1596/1813-9450-8024

Ara, S., and Shrivastav, P. K. (2025). Decoding rising female labour force participation in India: Evidence from PLFS data. *Economic and Political Weekly*, *60*(26–27). https://doi.org/10.71279/epw.v60i26-27.39982

Bardasi, E., Beegle, K., Dillon, A., and Serneels, P. (2011). Do labor statistics depend on how and to whom the questions are asked? Results from a survey experiment in Tanzania. *The World Bank Economic Review*, *25*(3), 418–447. https://doi.org/10.1093/wber/lhr022

Beneria, L. (1981). Conceptualizing the labor force: The underestimation of women's economic activities. *The Journal of Development Studies*, *17*(3), 10–28. https://doi.org/10.1080/00220388108421795

Chatterjee, E., Desai, S., and Vanneman, R. (2018). Indian paradox: Rising education, declining women's employment. *Demographic Research*, *38*, 855–878. https://doi.org/10.4054/DemRes.2018.38.31

Desai, S., and Joshi, O. (2019). The paradox of declining female work participation in an era of economic growth. *The Indian Journal of Labour Economics*, *62*(1), 55–71. https://doi.org/10.1007/s41027-019-00162-z

Deshpande, A., and Kabeer, N. (2024). Norms that matter: Exploring the distribution of women's work between income generation, expenditure-saving and unpaid domestic responsibilities in India. *World Development*, *174*, 106435. https://doi.org/10.1016/j.worlddev.2023.106435

Gaddis, I., and Klasen, S. (2014). Economic development, structural change, and women's labor force participation: A reexamination of the feminization U hypothesis. *Journal of Population Economics*, *27*(3), 639–681. https://doi.org/10.1007/s00148-013-0488-2

Goldar, B., and Aggarwal, S. C. (2026). Is the productivity of unpaid female family workers in Indian agriculture very low? *Economic and Political Weekly*, *61*(5). https://doi.org/10.71279/epw.v61i5.46716

Goldin, C. (1994). *The U-shaped female labor force function in economic development and economic history* (NBER Working Paper 4707). National Bureau of Economic Research.

https://doi.org/10.3386/w4707

Hirway, I., and Jose, S. (2011). Understanding women's work using time-use statistics: The case of India. *Feminist Economics*, *17*(4), 67–92. https://doi.org/10.1080/13545701.2011.622289

International Labour Organization. (2013). *Resolution concerning statistics of work, employment and labour underutilization, adopted by the nineteenth international conference of labour statisticians*. Geneva: ILO. https://www.ilo.org/resource/resolution-concerning-statistics-work-employment-and-labour-0

Jayachandran, S. (2021). Social norms as a barrier to women's employment in developing countries. *IMF Economic Review*, *69*(3), 576–595. https://doi.org/10.1057/s41308-021-00140-w

Jha, M., Ghatak, M., and Singh, J. (2026). Rise of unpaid family workers: Evidence on distress-driven employment growth in India. *Economic and Political Weekly*, *61*(1). https://doi.org/10.71279/epw.v61i01.46999

Kitagawa, E. M. (1955). Components of a difference between two rates. *Journal of the American Statistical Association*, *50*(272), 1168–1194. https://doi.org/10.1080/01621459.1955.10501299

Klasen, S., and Pieters, J. (2015). What explains the stagnation of female labor force participation in urban India? *The World Bank Economic Review*, *29*(3), 449–478. https://doi.org/10.1093/wber/lhv003

Lahoti, R., and Swaminathan, H. (2016). Economic development and women's labor force participation in India. *Feminist Economics*, *22*(2), 168–195. https://doi.org/10.1080/13545701.2015.1066022

Langsten, R., and Salen, R. (2008). Two approaches to measuring women's work in developing countries: A comparison of survey data from Egypt. *Population and Development Review*, *34*(2), 283–305. https://doi.org/10.1111/j.1728-4457.2008.00220.x

Lumley, T. (2004). Analysis of complex survey samples. *Journal of Statistical Software*, *9*(8). https://doi.org/10.18637/jss.v009.i08

Mehrotra, S., and Parida, J. K. (2017). Why is the labour force participation of women declining in India? *World Development*, *98*, 360–380. https://doi.org/10.1016/j.worldd

ev.2017.05.003

National Sample Survey Office. (2016). *Periodic labour force survey: Instructions to field staff, volume I: Design, concepts, definitions and procedures*. New Delhi: Ministry of Statistics and Programme Implementation, Government of India. Released with the unit-level data for July 2017 – June 2018.

National Sample Survey Office. (2023a). *Periodic labour force survey: Instructions to field staff, volume I: Design, concepts, definitions and procedures*. New Delhi: Ministry of Statistics and Programme Implementation, Government of India. Released with the unit-level data for July 2023 – June 2024.

National Sample Survey Office. (2023b). *Updated instruction to field staff of periodic labour force survey (PLFS) for July 2023 – June 2024*. New Delhi: Ministry of Statistics and Programme Implementation, Government of India. Released with the unit-level data.

National Sample Survey Office. (2024). *Press note on periodic labour force survey (PLFS) annual report [July 2023 – June 2024]*. New Delhi: Ministry of Statistics and Programme Implementation, Government of India, 23 September. https://www.mospi.gov.in/sites/default/files/press_release/Press_note_AR_PLFS_2023_24_22092024.pdf

Sarkar, S., Sahoo, S., and Klasen, S. (2019). Employment transitions of women in India: A panel analysis. *World Development*, *115*, 291–309. https://doi.org/10.1016/j.worlddev.2018.12.003

Appendix

Table A1: Comparison of field instructions relevant to measuring women's work, PLFS 2017–18 and 2023–24

Element	2017–18	2023–24
Definition of economic activity: primary production for own consumption, including free collection	Included	Included
Codes 92 (domestic duties only) and 93 (domestic duties and free collection etc. for household use)	Included	Included
Subsidiary status: economic activity for 30 days or more in the reference year	Included	Included
List of activities needing special care: kitchen garden, orchards, household poultry, dairy, free collection of fish, small game, wild fruits, vegetables for household consumption or sale	Not included (list has items a–h)	Added as item (j) of Box 8
Worked examples distinguishing codes 92/93 from subsidiary self-employment (usual status)	Not included	Added (Box 9)
Worked examples for codes 92/93 in the daily record (current weekly status)	Not included	Added (Box 10)
Question on destination of output of self-employed in NIC 01–09	Not asked	Asked

Source: Instructions to Field Staff, Volume I, for 2017–18 and 2023–24 (Chapter Three), and the note on updated instructions for 2023–24, which dates the clarifications in Boxes 9 and 10 to PLFS 2022–23.

Table A2: Robustness: alternative decomposition cells and working ages

Panel A: Kitagawa decomposition, alternative cells	Cells	Unmatched (%)	Composition	Rates
Cells: sector	2	0.00	0.06	18.30
Cells: state × sector	70	0.00	-0.06	18.42
Cells: sector × age × education	40	0.00	-0.69	19.06
Cells: state × sector × age × education × marital status	4,093	0.32	-1.06	19.42
Cells: state × sector × age × education × marital status × social group	10,625	1.26	-0.41	18.78
Panel B: women aged 15–59	*2017–18*	*2023–24*	*Change*	*SE*
Worker-population ratio	23.8	43.7	19.9	0.38
Own-account	4.7	13.3	8.7	0.22
Unpaid helper	7.5	15.9	8.3	0.27
Regular wage	5.2	7.3	2.2	0.16
Casual	6.4	7.1	0.7	0.20
Subsidiary only	2.4	11.8	9.4	0.24

Panel A: change in women's WPR (all India, 15+, percentage points) decomposed with alternative cell definitions; unmatched is the population share in cells present in one year only. Panel B: usual status (ps+ss), per cent of women aged 15–59; change and its standard error in percentage points.

Table A3: Change in women's worker-population ratio and its components, 20 most populous states, 2017–18 to 2023–24 (percentage points)

State	Worker-population ratio	Own-account	Unpaid helper	Wage (regular and casual)
Assam	37.2	22.2	11.8	3.2
Jharkhand	35.0	12.0	21.1	1.9
Odisha	29.7	10.3	15.5	4.0
Gujarat	26.5	10.4	10.6	5.6
Bihar	26.1	10.6	13.3	2.3
Rajasthan	22.5	7.7	10.7	4.1
Madhya Pradesh	20.9	7.9	14.0	-0.9
Uttar Pradesh	20.5	9.8	9.7	1.0
Jammu and Kashmir	19.5	23.2	-4.0	0.3
West Bengal	19.0	12.3	6.1	0.6
Kerala	15.7	5.2	2.2	8.4
Punjab	15.2	7.3	2.1	5.8
Telangana	14.0	3.8	6.5	3.8
Karnataka	12.4	3.3	5.9	3.2
Haryana	10.8	3.6	4.3	2.9
Chhattisgarh	10.4	0.6	6.8	3.0
Tamil Nadu	10.2	4.1	0.4	5.7
Maharashtra	10.0	3.9	2.7	3.3
Delhi	5.5	0.8	1.2	3.4
Andhra Pradesh	2.3	0.7	2.9	-1.3

Usual status (ps+ss), women aged 15 and over. States ordered by the change in the worker-population ratio; the three components sum to it. Jammu and Kashmir includes Ladakh.